Basics of Graph Theory

Dr. Sayantan Panja

1 Introduction

Let there are five cities and seven highways connecting them as shown in Fig. 1.1(a) or Fig. 1.1(b).

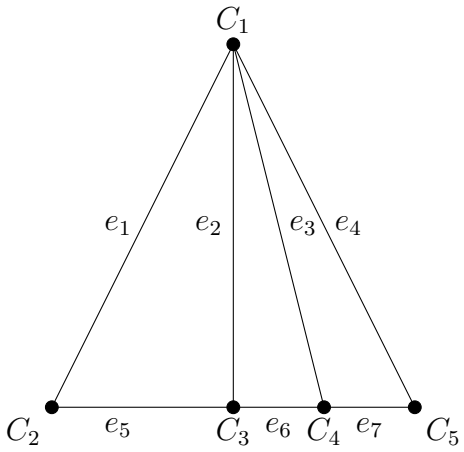


Fig. 1.1(a)

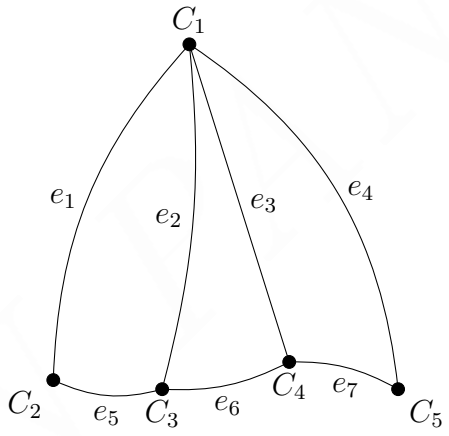


Fig. 1.1(b)

Let the points C_1, C_2, C_3, C_4, C_5 represent the five cities and $e_1, e_2, e_3, e_4, e_5, e_6, e_7$ are the seven highways connecting these cities.

Here points (or dots) represent the cities and the lines connecting these points represent the highways. These lines are not necessarily straight lines. In Fig. 1.1(a) all are straight lines but in Fig. 1.1(b) some are curve lines. This diagram of Fig. 1.1(a) or Fig. 1.1(b) is known as **Graph** of this network of cities and highways. The points (or dots) $C_1, C_2, \cdots, C_5$ are known as **vertices** and the lines (not necessarily straight lines) joining any two vertices are known as **edges**. Here $e_1, e_2, \cdots, e_7$ are edges. It is noted that a vertex without edge is possible but an edge without vertex is not possible.

2 Graphs and Subgraphs

Definition 2.1. *A graph G is a pair (V, E) where V is a finite non-empty set of vertices and E is a finite set (may be empty) of edges such that each edge $e \in E$ is assigned an unordered pair of vertices (u, v); $u, v \in V$. If e is assigned an unordered pair of vertices (u, v), then u and v are called the end vertices of e and in this case we write $e = uv$.*

When a vertex v is an end vertex of some edge e, we say that e is incident with the vertex v and v is incident with the edge e.

Example 2.2. *Let $V = \{v_1, v_2, v_3, v_4\}$ and $E = \{e_1, e_2, e_3, e_4, e_5, e_6\}$. The edges are given by $e_1 \leftrightarrow (v_1, v_2)$; $e_2 \leftrightarrow (v_3, v_4)$; $e_3 \leftrightarrow (v_1, v_3)$; $e_4 \leftrightarrow (v_2, v_4)$; $e_5 \leftrightarrow (v_1, v_2)$; $e_6 \leftrightarrow (v_2, v_4)$. Then $G = (V, E)$ is a graph. The graph may be represented by the following diagram:*

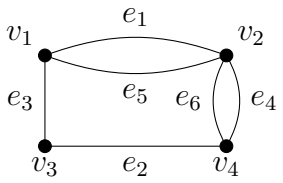


Fig. 2.1

Definition 2.3. *Vertices are also called points, nodes or just dots. If e is an edge with end vertices u and v, then e is said to join u and v. Let e be an edge. If the end vertices of e are identical, then e is said to be a loop.*

Example 2.4. *Consider the following graph. Here e_1 is a loop.*

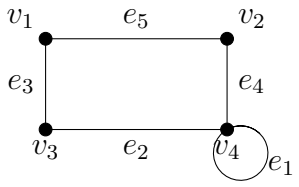


Fig. 2.2

Definition 2.5. *Let G be a graph. If two (or more) edges of G are associated with the same pair of vertices, then these edges are called **parallel edges**.*

Example 2.6. *Consider the graph Fig 2.1. In this graph the edges e_4 and e_6 are parallel. Also e_1 and e_5 are parallel edges.*

Definition 2.7. *A graph G is said to be a **simple graph** if it has no loops and no parallel edges.*

Example 2.8. *The graph in Fig 2.2 does not contain any parallel edges but it contains a loop. Therefore this graph is not a simple graph.*

Definition 2.9. *Let G be a graph and v be a vertex of G. The **degree** of v, denoted by $d(v)$, is the number of edges of G incident with the vertex v.*

We take the convention that each loop on a vertex v contributes 2 to the degree of v. So, degree of any loop is 2. A vertex is called an even degree vertex or odd degree vertex if the degree of this vertex is respectively even or odd.

Definition 2.10. *Let G be a graph. A vertex of G is said to be a **pendent** vertex, if it is of degree 1; and an **isolated** vertex if it is of degree 0. That is, a vertex which is incident with no edge, is called isolated vertex.*

Definition 2.11. *If every vertex of a graph is isolated, then it is called **null graph**.*

Theorem 2.12 ((First Theorem of Graph Theory))**.** *The sum of the degrees of all vertices of a graph is an even integer.*

Proof. Let G be a graph with n edges. Each edge e is either a loop or incident with two distinct vertices. If e is a loop on a vertex v, then e contributes 2 to the sum of the degrees of v. On the other hand if e is incident with two distinct vertices v and u, then e contributes 1 to the degrees of each vertex. Thus we find that, when we take the sum of the degrees of the vertices, each edge contributes 2 to this sum. Hence the sum of the degrees of the vertices is twice the number of edges. This implies that the sum of the degrees of the vertices is $2n$, which is an even integer. □

Theorem 2.13. *The number of vertices of odd degree in a graph is always even number.*

Proof. Suppose a graph G has n odd degree vertices $v_1, v_2, \cdots, v_n$ and m even degree vertices $u_1, u_2, \cdots, u_m$. Then by Theorem 2.12 we have,

$$\sum_{i=1}^{n} d(v_i) + \sum_{i=1}^{m} d(u_i) = \text{an even integer.}$$

Since each $d(u_i)$ is even, so it follows that $\sum_{i=1}^{m} d(u_i)$ is an even integer. Then, $\sum_{i=1}^{n} d(v_i)$ is an even integer. Since, each $d(v_i)$ is odd integer, therefore n must be even. This completes the proof. □

Example 2.14. *Draw a graph having the given properties or explain why no such graph exists.*

(a) Simple graph with four vertices each of degree two.

(b) Graph with four edges, four vertices having degrees 1, 2, 3, 4.

Solution.

(a) A graph satisfying the given properties is

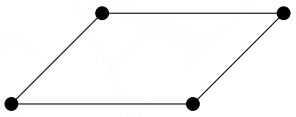


Fig. 2.2

(b) Suppose there exists a graph G with the given properties. Then the sum of the degrees of the vertices of G is 10. In any graph we know that the sum of the degrees of all vertices is twice the number of edges. Hence G must have five edges, which is a contradiction. Thus, there does not exist such graph.

Example 2.15. *Does there exist a simple graph with five vertices having degrees* $2, 2, 4, 4, 4$*? Justify.*

Solution. Suppose there exists a simple graph with five vertices v_1, v_2, v_3, v_4, v_5 having degrees $2, 2, 4, 4, 4$ respectively. Since the graph does not contain any loop and any parallel edges, it follows that each of the vertices v_3, v_4, v_5 must have four adjacent vertices. Hence v_1 is an adjacent vertex of v_3, v_4, v_5. This implies that the degree of v_1 is at least 3, which goes against the assumption. Hence there is no such graph.

Definition 2.16. *A graph consisting finite number of vertices is called* ***finite graph****.*

Theorem 2.17. *Let G be a simple graph with n-vertices. Then, degree of any vertex can not exceed $(n-1)$, and number of edges can not exceed $\frac{n(n-1)}{2}$.*

Proof. Let, v be a vertex of a simple graph G and $d(v) = k$. Then, there are k-number of edges incident with v. Since, G is a simple graph, so there are no parallel edge and no loop. So, the other vertices of these k-edges are distinct. Therefore, there exists at least k-number of vertices, other than v. That is, $n \geq k-1$, implies $k \leq n-1$.

Let, $v_1, v_2, \cdots v_n$ be the n-vertices of G and e be the number of edges of G. Then, $\sum_{i=1}^{n} d(v_i) = 2e$. Again, $d(v_i) \leq n-1$, for each $i = 1, 2, \cdots n$.

Therefore, $2e = \sum_{i=1}^{n} d(v_i) \leq \sum_{i=1}^{n} (n-1) = n(n-1)$, implies, $e \leq \frac{n(n-1)}{2}$. □

Theorem 2.18. *Every simple graph with $n \geq 2$ vertices has at least two vertices with the same degree.*

Proof. Let G be a simple graph with n vertices. In a simple graph there are no loop, no parallel edge. So, the possible degrees of each vertex is $0, 1, 2, \cdots, n-1$.

Claim: In a simple graph, it is impossible to have both a vertex of degree 0 and another of degree $n-1$ simultaneously.

Suppose, there is one vertex of degree 0 (i.e., an isolated vertex).

If there is another vertex of degree $n-1$, then it must be connected to all other $n-1$ vertices, which is a contradiction since the isolated vertex cannot be connected to it.

Hence, our claim is justified.

Therefore, the actual set of possible degrees of each vertex is either $\{0, 1, 2, \ldots, n-2\}$ or $\{1, 2, \ldots, n-1\}$, which means there are at most $n-1$ distinct possible degree.

Now, since there are n vertices and only $n-1$ possible degree values, by the *Pigeonhole Principle*, at least two vertices must have the same degree. □

Definition 2.19. *A simple graph in which there exists an edge between every pair of vertices is said to be a* ***complete graph*** *or clique.*

Complete graph with n-vertices is denoted by K_n.

Example 2.20. *The following graphs are the examples of complete graph.*

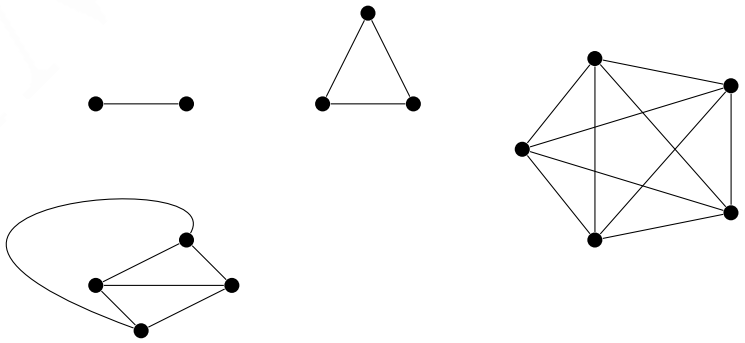


Fig. 2.3

Note 2.21. *In a complete graph with n-vertices, each vertex has the same degree which is $(n-1)$.*

Theorem 2.22. *A complete graph with n-vertices contains $\frac{n(n-1)}{2}$ edges.*

Proof. Let, G be a complete graph with n-vertices. Due to completeness of the graph, every vertex is connected to $(n-1)$ other vertices. As there are n-vertices, so it seems there are total $n(n-1)$ edges. But there is a double count. After removing the double count, total number of edges is $\frac{n(n-1)}{2}$. □

Definition 2.23. *Let $G = (V, E)$ and $G' = (V', E')$ be two graphs. Then G' is said to be a subgraph of G if $V' \subseteq V$ and $E' \subseteq E$.*

Example 2.24. *Consider the following graphs:*

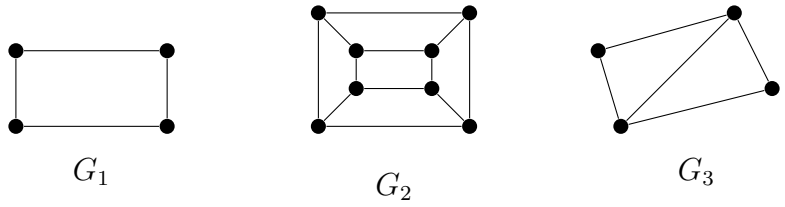


Fig. 2.4

Here, G_1 is a subgraph of both G_2 and G_3, but G_3 is not a subgraph of G_2.

Definition 2.25. *Two graphs G and G' are said to be **isomorphic** (or equivalent), if there exists a one-one correspondence between their vertices and between their edges such that the incidence relationship is preserved.*
In other words, two graphs G and G' are said to be isomorphic if G and G' have,

(i) same number of vertices.

(ii) same number of edges.

(iii) the corresponding vertices have the same degree.

(iv) the incidence relationship between vertices and edges are preserved.

Example 2.26. *Show that the following graphs are isomorphic.*

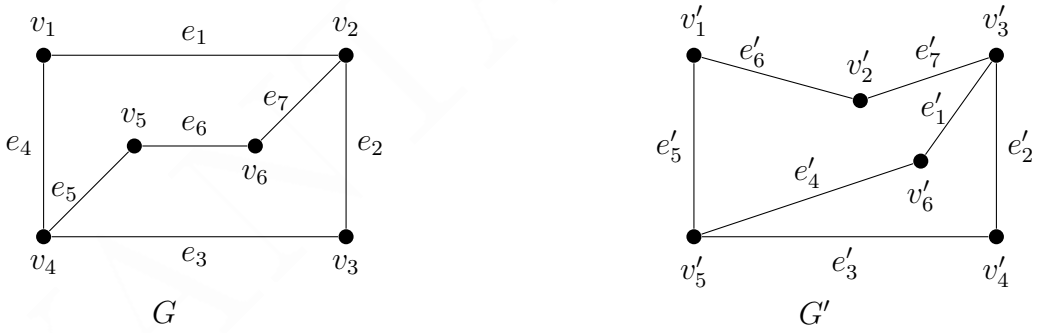


Fig. 2.5

Solution. The vertices v_1, v_2, v_3, v_4, v_5 and v_6 corresponds to $v'_6, v'_3, v'_4, v'_5, v'_1$ and v'_2 respectively. The edges $e_1, e_2, e_3, e_4, e_5, e_6$ and e_7 corresponds to $e'_1, e'_2, e'_3, e'_4, e'_5, e'_6$ and e'_7.

Again, $d(v_1) = d(v'_6) = 2$, $d(v_2) = d(v'_3) = 3$, $d(v_3) = d(v'_4) = 2$, $d(v_4) = d(v'_5) = 3$, $d(v_5) = d(v'_1) = 2$, $d(v_6) = d(v'_2) = 2$.

Hence, G and G' are isomorphic.

Example 2.27. *Show that the following graphs are isomorphic.*

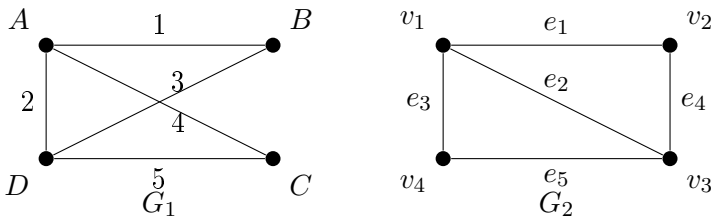


Fig. 2.6

Example 2.28. *Check whether the following graphs are isomorphic or not.*

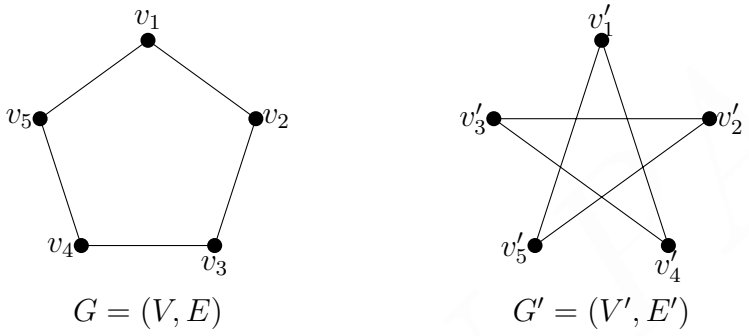


Fig. 2.7

Definition 2.29. *A graph $G = (V, E)$ is called a* ***bipartite graph*** *if the vertex set V of G can be partitioned into two nonempty subsets V_1 and V_2 such that each edge of G is incident with one vertex of V_1 and one vertex in V_2.*

A bipartite graph G is called a *complete bipartite graph on m and n vertices* if G is simple and if the vertex set V of G can be partitioned into two nonempty subsets V_1 and V_2 with m and n vertices respectively such that there is an edge between each pair of vertices v_1 and v_2 where $v_1 \in V_1$ and $v_2 \in V_2$. A complete bipartite graph on m and n vertices is denoted by $K_{m,n}$.

Example 2.30. *The complete bipartite graph on 2 and 3 vertices ($K_{2,3}$) and complete bipartite graph on 3 and 4 vertices ($K_{3,4}$) are shown in the following figures:*

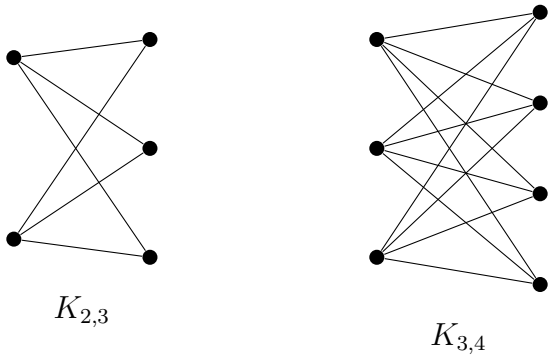


Fig. 2.8

3 Walks, Paths, Circuits, Connected & Disconnected Graph

Definition 3.1. *Let u and v be two vertices in a graph G. A **walk** of length n from u to v in G is an alternating sequence of $(n+1)$ vertices and n edges of G $(u = v_1, e_1, v_2, e_2, \cdots, v_n, e_n, v_{n+1} = v)$ beginning with vertex u and ending with vertex v.*
A walk is called closed walk if $u = v$, and open walk if $u \neq v$.

Definition 3.2. *A walk with no repeated edges is called a **trail** and a walk with no repeated vertices is called a **path**.*

Remark 3.3. *A path is a walk in which no vertex is repeated. Thus, in a path no edge can be repeated. So every path is a trail. But the converse is not true. It follows from the following example.*

Example 3.4. *Consider the following graph:*

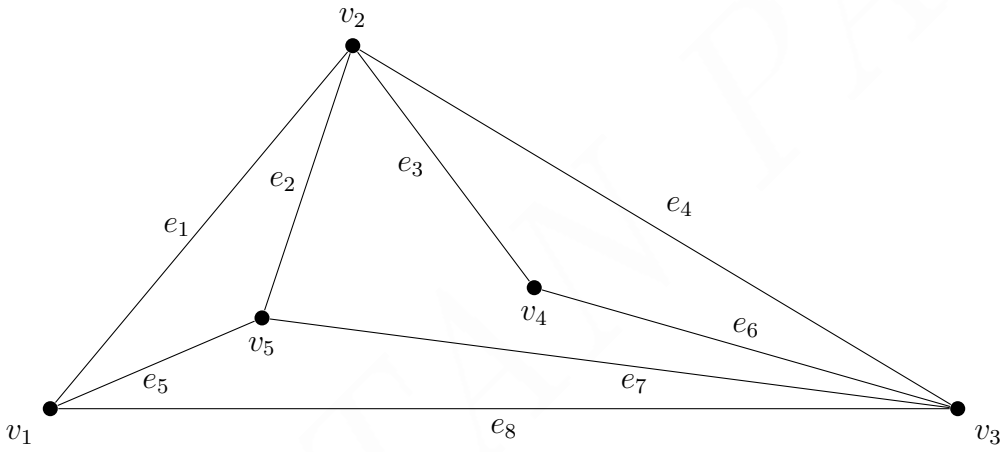


Fig. 3.1

In this graph, the walk $(v_1, e_5, v_5, e_2, v_2, e_3, v_4, e_6, v_3, e_8, v_1)$ is a trail but not a path.

Definition 3.5. *A trail of non-zero length from a vertex u to itself is called a **circuit**.*

Example 3.6. *In the graph given in Fig. 3.1, the trail $(v_1, e_5, v_5, e_2, v_2, e_3, v_4, e_6, v_3, e_8, v_1)$ is a circuit.*

Remark 3.7. *From definition, it follows, a circuit is a closed walk of non-zero length from a vertex u to itself with no repeated edge.*

Definition 3.8. *A trail of non-zero length from a vertex v to itself in a graph is called a **cycle**, if there is no repeated vertex except the vertex v (beginning and end vertex).*

Example 3.9. *In the graph given in Fig. 3.1, the walk $(v_1, e_5, v_5, e_2, v_2, e_3, v_4, e_6, v_3, e_8, v_1)$ is a cycle.*

Remark 3.10. *From definition, it follows, every cycle is a circuit but the converse is not true in general.*

Example 3.11. *In the graph given in Fig. 3.1, the trail $(v_1, e_5, v_5, e_2, v_2, e_3, v_4, e_6, v_3, e_4, v_2, e_1, v_1)$ is a circuit but not a cycle.*

Definition 3.12. *Let G be a graph and u, v be two vertices in G. Then G is said to be connected graph if there is a path from u to v. Otherwise, G is said to be disconnected graph.*

Example 3.13. *The graph in Fig. 3.1 is connected graph, whereas the following graph is disconnected graph.*

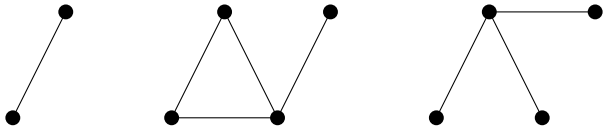


Fig. 3.2

Definition 3.14. *Let $G = (V, E)$ be a graph and $u \in V$. Let, $C(u)$ denotes the set of all vertices in G that are connected to u. i.e.,*

$$C(u) := \{v \in V : \text{ there is a path from } v \text{ to } u\}.$$

Here, $C(u)$ is called the connected component of the graph G containing u.

Example 3.15. *The graph given in Fig. 3.2 is a graph with three connected components.*

Theorem 3.16. *A simple graph with n vertices and m components can have at most $\frac{(n-m)(n-m+1)}{2}$ edges.*

4 Eulerian Graph

Definition 4.1. *A circuit in a graph that includes all the edges of the graph is called an **Euler circuit**.*

Example 4.2. *Consider the following graph:*

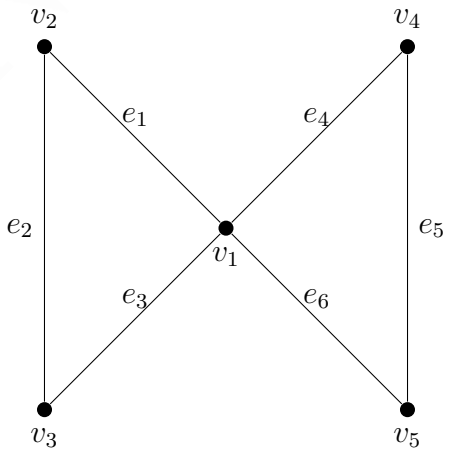


Fig. 4.1

In this graph $(v_1, e_1, v_2, e_2, v_3, e_3, v_1, e_4, v_4, e_5, v_5, e_6, v_1)$ is an Euler circuit.

Definition 4.3. *A graph G is said to be **Eulerian** if G is null graph (i.e., only one vertex and no edge) or G has an Euler circuit.*

The Königsberg Bridge Problem

The city of Königsberg (now called Kaliningrad, located in Russia) was set on both sides of the Pregel River and included two large islands A and D connected to each other and to the mainland B and C with seven bridges as shown in the Fig. 4.2. For some purpose (maintenance or etc.) the problem was faced that:

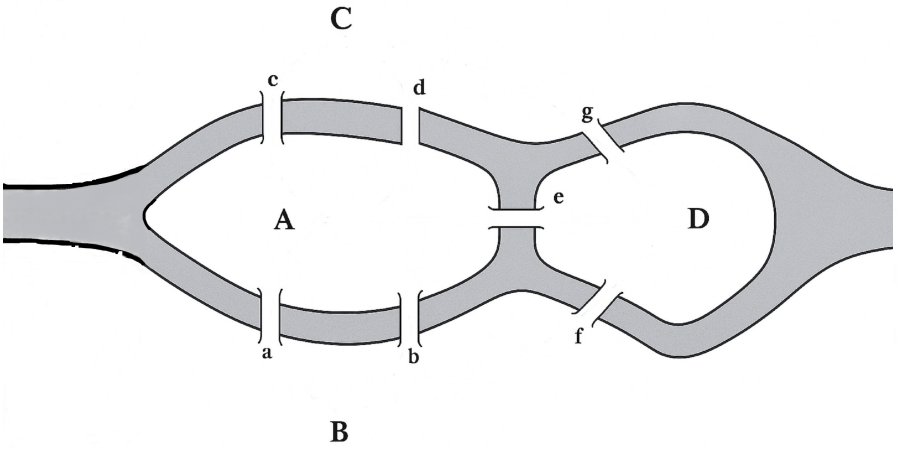


Fig. 4.2

starting from any one of the four land areas (A, B, C and D) walk over each of the seven bridges exactly once and return to the starting point.

Leonhard Euler represented this problem using the concept of a graph, in which each land area (the two islands and the two riverbanks) is represented as a vertex, and each bridge is represented as an edge connecting two vertices, as shown in the Fig. 4.3.

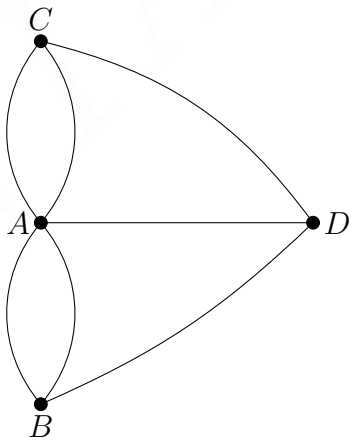


Fig. 4.3

Thus, in graph theory, the problem was reformulated as:

> "Given a graph, is it possible to find a closed trail (starting and ending at the same vertex) that uses every edge exactly once?"

This problem is known as **Konisgberg Bridge Problem** and this Problem is historically regarded as the first theorem in graph theory.

In 1736, Euler proved that a solution for this problem does not exist (see the next theorem). That is,

> *"It is not possible to walk over each of the seven bridges exactly once and return to the starting point."*

Theorem 4.4. *If a connected graph G is Eulerian, then every vertex of G has even degree.*

Proof. Suppose the graph G is Eulerian. If G has one vertex v and no edges, then $d(v) = 0$ which is even. Assume that the graph G contains more than one vertex. Since G is Eulerian, G has an Euler circuit; say

$$C = (v_1, e_1, v_2, e_2, \ldots, e_n, v_{n+1} = v_1).$$

Let u be a vertex of G. Since G is a connected graph, u is not an isolated vertex and hence u is an end vertex of some edge. Since C contains all the edges, so u is a member of C. If $u = v_1$, then u is also v_{n+1} and the edge e_1 and e_n contribute 2 towards the degree of u. Suppose u is not v_1. Then each appearance (if any) of u other than the first and the last appearance in C, contributes 2 towards the degree of u. Again, C contains all the vertices and all the edges of G, but there are no repeated edges in C though a vertex may appear more than once. Hence it follows that the degree of u is even. □

Note 4.5. *The graph in Fig. 4.3 contains odd degree vertices. Hence the graph has no Euler circuit. This gives the solution of the Konigsberg bridge problem.*

5 Hamiltonian Graph

Definition 5.1. *A cycle in a graph G is called a **Hamiltonian cycle** if it contains each vertex of G. If a graph has a Hamiltonian cycle, then the graph is called a **Hamiltonian graph**.*

Note 5.2. *From definition it follows that, each vertex in the Hamiltonian cycle (except the first/last which are the same) must be connected to two distinct edges (entering and leaving) in the cycle. i.e., each vertex in the cycle C is of degree 2.*

Therefore, a Hamiltonian graph must contain a cycle in which each vertex degree is 2.

Example 5.3. *Consider the following graph:*

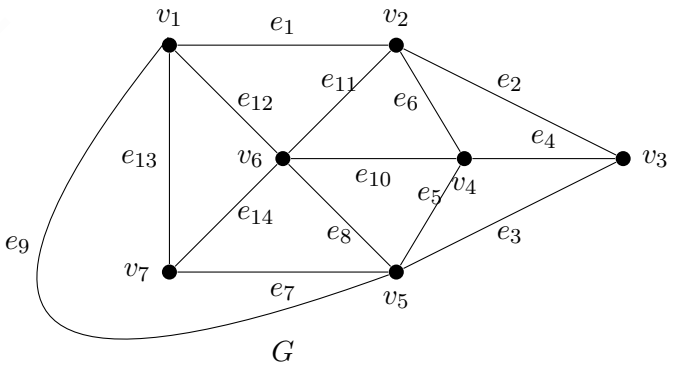


Fig. 5.1

In this graph, $C = (v_1, e_1, v_2, e_2, v_3, e_3, v_5, e_5, v_4, e_{10}, v_6, e_{14}, v_7, e_{13}, v_1)$ is a cycle, which contains all the vertices of G. So this cycle is a Hamiltonian cycle and hence G is a Hamiltonian graph.

Note that, this graph G is **not** Eulerian, as G contains odd degree vertices.

Example 5.4. *To show, an Eulerian graph need not necessarily a Hamiltonian graph, we consider the following example of the graph.*

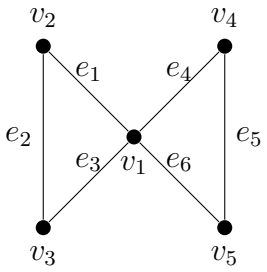


Fig. 5.2

In Fig. 5.2, the graph G is connected, with each vertex is of even degree. Hence G is an Eulerian graph. We now show that G is not Hamiltonian. Suppose, G is a Hamiltonian graph. Then, G has a Hamiltonian cycle C. Then, C must contain the vertex v_1. As in a Hamiltonian cycle each vertex is of degree 2, so $d(v_1) = 2$, which is a contradiction as in the Fig 5.2, it shows that $d(v_1) = 4$. Hence G has no Hamiltonian cycle.

Example 5.5. *The graph G_1 in the Fig. 2.4 is both Eulerian as well as Hamiltonian graph.*

6 Digraph

Definition 6.1. *A directed graph (or a Digraph, in short) consists of vertex set V and edge set E such that each edge $e \in E$ is associated with an ordered pair of vertices.*

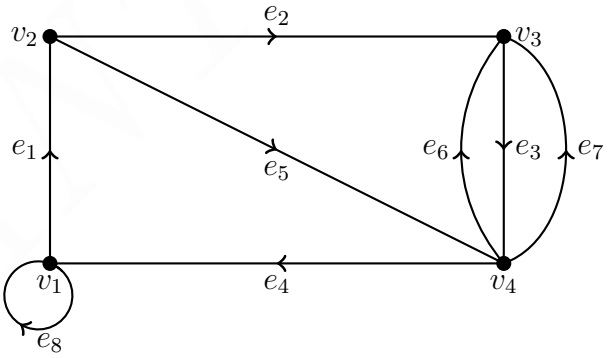


Fig. 6.1

In the Fig. 6.1, $V = \{v_1, v_2, v_3, v_4\}$ and $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$. Here the edge e_1 corresponds to the ordered pair (v_1, v_2), but not (v_2, v_1). That is $e_1 = (v_1, v_2)$ but $e_1 \neq (v_2, v_1)$. We us the arrow sign on each edge to show the ordering.

Similarly, $e_2 = (v_2, v_3)$, $e_3 = (v_3, v_4)$, $e_4 = (v_4, v_1)$, $e_5 = (v_2, v_4)$, $e_6 = (v_4, v_3) = e_7$, and $e_8 = (v_1, v_1)$.

Note 6.2. *1. The graph where every edge has a definite direction is called digraph.*

2. *In a digraph, we do not say an edge is incident to a vertex. Rather we say, an edge is* ***incident out*** *and* ***incident into*** *of a vertex. For example, in Fig. 6.1, e_4 is incident into v_1 and incident out v_4. The edge e_8 is incident in and incident out v_1.*

3. *In Fig. 6.1, edges e_6 and e_7 are parallel but e_3 and e_6 are not parallel in the digraph, as their directions are different.*

Definition 6.3. *In a digraph, the number of edges incident out of a vertex v is called* ***out-degree*** *of v and is denoted by $deg^+(v)$.*

In a digraph, the number of edges incident into of a vertex v is called ***in-degree*** *of v and is denoted by $deg^-(v)$.*

For example, in Fig. 6.1, $deg^-(v_3) = 3$ and $deg^+(v_3) = 1$, $deg^-(v_1) = 2$ and $deg^+(v_1) = 2$.

Note 6.4. *In a digraph, for any vertex v, $deg(v) = deg^-(v) + deg^+(v)$.*